POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6023A | Semester VI | Theory | 4 Credits

Logistics and Supply Chain Management

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6023A as Logistics and Supply Chain Management in Semester VI for Mechanical Engineering. The official SITTTR detailed course page was unavailable. With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Logistics & Supply Chain Management course and NPTEL IIT Madras’s Operations and Supply Chain Management course. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Worked examples are for education; purchasing, routing, staffing, contractual, regulatory, safety and commercial decisions require current data, authorised systems, applicable law and competent review.

Verified source note: Verified sources support logistics/SCM concepts, warehousing, KPIs, reverse/green/lean logistics, supply-chain drivers and strategy, sourcing, distribution, forecasting, transport, inventory, digitalisation, resilience and sustainability. The four-module presentation is a study sequence, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Logistics and Supply Chain Management studies the planning and coordination of material, information and financial flows that connect suppliers, operations, distribution channels and customers. It develops systems thinking across warehousing, transport, inventory, sourcing, forecasting, network design, performance and sustainability.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Basic arithmetic and graphs, business/industrial awareness, operations and manufacturing fundamentals, spreadsheet familiarity and ethical handling of operational data.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain logistics, supply-chain structure, stakeholders and the relationship between logistics and SCM.
2. Explain warehousing, inventory, transportation, distribution and network-design decisions.
3. Explain forecasting, planning, sourcing, coordination and supply-chain performance concepts.
4. Explain digital, resilient, sustainable and reverse-logistics approaches using responsible decision-making.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□ exam-□ □□□□□ □□□□□□□ □□□□□□ □□□□□ □□□□□□□□.
> Define, explain, apply □□□□□ action words □□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain logistics functions, supply-chain flows, stakeholders and value creation.	Understand
CO2	Explain warehousing, inventory, transportation and distribution-network choices.	Understand
CO3	Apply basic forecasting, planning, sourcing, coordination and performance-measurement concepts to a case.	Apply
CO4	Explain digitalisation, sustainability, resilience and reverse logistics in modern supply chains.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Logistics and Supply-Chain Foundations	Logistics definition and evolution; SCM networks; product, information and financial flows; stakeholders; service, cost and value; manufacturing and service contexts.	Approx. 14 hours	CO1	Module 1
2	Warehousing, Inventory, Transportation and Distribution	Storage and warehouse roles; inventory costs and policies; transportation modes and routing; distribution networks, facility location and service design.	Approx. 16 hours	CO2	Module 2
3	Planning, Sourcing, Coordination and Performance	Demand forecasting, aggregate/capacity planning, scheduling awareness, make-or-buy, sourcing and vendor rating, coordination, KPIs and balanced performance views.	Approx. 16 hours	CO3	Module 3
4	Digital, Sustainable and Resilient Supply Chains	E-logistics, traceability and data; reverse, green and lean logistics; disruption risk, resilience, emergency response, circularity and responsible asset management.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Logistics and Supply-Chain Foundations

Logistics definition and evolution; SCM networks; product, information and financial flows; stakeholders; service, cost and value; manufacturing and service contexts.

CO1 · Approx. 14 hours

Warehousing, Inventory, Transportation and Distribution

Storage and warehouse roles; inventory costs and policies; transportation modes and routing; distribution networks, facility location and service design.

CO2 · Approx. 16 hours

Planning, Sourcing, Coordination and Performance

Demand forecasting, aggregate/capacity planning, scheduling awareness, make-or-buy, sourcing and vendor rating, coordination, KPIs and balanced performance views.

CO3 · Approx. 16 hours

Digital, Sustainable and Resilient Supply Chains

E-logistics, traceability and data; reverse, green and lean logistics; disruption risk, resilience, emergency response, circularity and responsible asset management.

CO4 · Approx. 14 hours

MODULE 1

Logistics and Supply-Chain Foundations

Logistics definition and evolution; SCM networks; product, information and financial flows; stakeholders; service, cost and value; manufacturing and service contexts.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

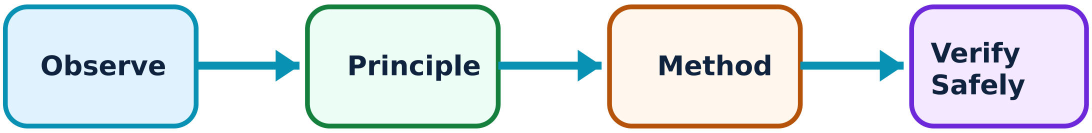
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Logistics and Supply-Chain Foundations: identify the system, state its principle, follow the correct method and verify the result safely.

1. From logistics to supply-chain management−

Logistics coordinates movement, storage and related information; supply-chain management coordinates a wider network of organisations and decisions from sourcing through operations, distribution and customer service. Performance depends on connected flows rather than one function in isolation.

Study and application method

Map a simple product journey from supplier to customer and label material, information and cash flows, hand-offs, lead times, ownership and likely service risks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map a simple product journey from supplier to customer and label material, information and cash flows, hand-offs, lead times, ownership and likely service risks.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A simplified map may omit compliance, labour, safety, finance and contractual conditions. State scope and assumptions before using it for analysis.

2. Stakeholders, objectives and trade-offs−

Customers, suppliers, manufacturers, service providers, regulators and communities can have different priorities. Decisions balance service level, cost, speed, reliability, capacity, risk, quality and environmental/social impact.

Study and application method

Use a stakeholder-and-objective table to identify a trade-off in a case, the metric each stakeholder values and the evidence needed before deciding.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

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Common mistake / precaution: Do not optimise a local metric such as transport cost if it transfers unacceptable risk, delay, waste or service failure elsewhere in the chain.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Warehousing, Inventory, Transportation and Distribution

Storage and warehouse roles; inventory costs and policies; transportation modes and routing; distribution networks, facility location and service design.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

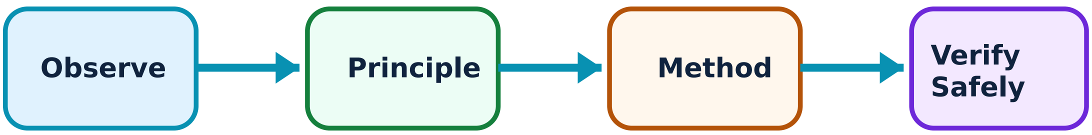
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Warehousing, Inventory, Transportation and Distribution: identify the system, state its principle, follow the correct method and verify the result safely.

1. Warehousing and inventory fundamentals−

Warehouses receive, store, protect, pick and dispatch goods while supporting consolidation, postponement and service availability. Inventory protects against uncertainty and supports flow, but it ties up capital and can create obsolescence, loss or handling risk.

Study and application method

Classify stock in a case (cycle, safety, seasonal or pipeline), identify relevant costs and data, then explain how lead time and demand variability influence the inventory policy.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Classify stock in a case (cycle, safety, seasonal or pipeline), identify relevant costs and data, then explain how lead time and demand variability influence the inventory policy.

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Common mistake / precaution: EOQ, reorder-point and safety-stock examples rely on assumptions. Use actual demand, service, shelf-life, capacity and policy data before operational use.

2. Transport and distribution-network design−

Transport modes, consolidation, routes, distribution-centre placement and last-mile arrangements affect cost, speed, capacity, emissions and resilience. Distribution networks must align with product characteristics and customer-service needs.

Study and application method

Compare two distribution alternatives with a transparent table covering distance, mode, lead time, handling points, capacity, service level, risk, emissions and information gaps.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare two distribution alternatives with a transparent table covering distance, mode, lead time, handling points, capacity, service level, risk, emissions and information gaps.

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Common mistake / precaution: Routing and facility recommendations need accurate network, vehicle, labour, regulatory and safety data; educational cases are not dispatch instructions.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Planning, Sourcing, Coordination and Performance

Demand forecasting, aggregate/capacity planning, scheduling awareness, make-or-buy, sourcing and vendor rating, coordination, KPIs and balanced performance views.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

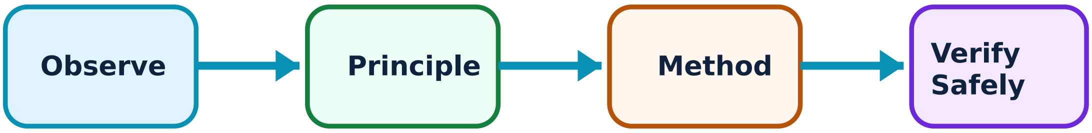
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Planning, Sourcing, Coordination and Performance: identify the system, state its principle, follow the correct method and verify the result safely.

1. Forecasting and operational planning−

Forecasts inform capacity, procurement, inventory and distribution plans but remain uncertain. Planning links demand pattern, lead time, capacity, constraints and service targets; it benefits from regular review and feedback.

Study and application method

Create a simple forecast-review worksheet: historical demand source, method assumption, forecast, error check, event adjustment, capacity implication and decision owner.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a simple forecast-review worksheet: historical demand source, method assumption, forecast, error check, event adjustment, capacity implication and decision owner.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Forecasts are estimates, not facts. Avoid using one short data series or an untested model as the sole basis for commitments or emergency stock decisions.

2. Sourcing, coordination and performance−

Make-or-buy, supplier selection, vendor rating and coordination affect continuity, cost, quality and risk. Useful KPIs balance service, cost, quality, assets, flexibility and sustainability rather than rewarding one silo alone.

Study and application method

Propose a small KPI dashboard for a case, defining each metric, calculation boundary, source, frequency, target rationale and unintended behaviour to watch for.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Propose a small KPI dashboard for a case, defining each metric, calculation boundary, source, frequency, target rationale and unintended behaviour to watch for.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Supplier scoring and contracts may involve confidential data, fairness, legal duties and strategic risk. Use authorised data and review processes.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Digital, Sustainable and Resilient Supply Chains

E-logistics, traceability and data; reverse, green and lean logistics; disruption risk, resilience, emergency response, circularity and responsible asset management.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

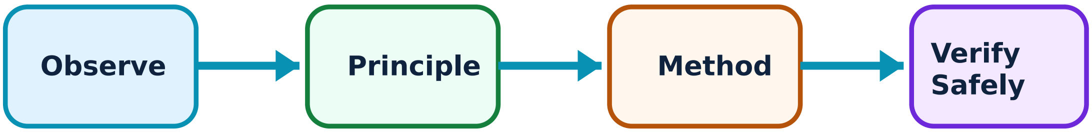
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Digital, Sustainable and Resilient Supply Chains: identify the system, state its principle, follow the correct method and verify the result safely.

1. Digitalisation and visibility−

Digital tools can improve visibility, planning, traceability and coordination when data quality, interoperability, cybersecurity and user workflows are addressed. Technology does not remove the need for governance and exception management.

Study and application method

Map a data flow for an order or shipment and identify data owner, update point, decision user, quality risk, access control and recovery action for missing/incorrect data.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map a data flow for an order or shipment and identify data owner, update point, decision user, quality risk, access control and recovery action for missing/incorrect data.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Operational data can be commercially sensitive and security-critical. Do not share, connect or automate systems without authorisation and appropriate controls.

2. Sustainability, reverse logistics and resilience−

Green logistics seeks lower environmental impact; reverse logistics manages returns, recovery, repair, reuse or disposal; resilience prepares networks to absorb and recover from disruption. These choices require a whole-life, whole-network view.

Study and application method

For a product-return case, identify return triggers, triage options, safety/quality checks, recovery route, transport/packaging needs, data capture and measures of environmental/service impact.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or

conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For a product-return case, identify return triggers, triage options, safety/quality checks, recovery route, transport/packaging needs, data capture and measures of environmental/service impact.

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Common mistake / precaution: Never assume returned goods are safe or reusable. Regulated, hazardous, damaged or personal-data-bearing goods need authorised handling and disposition processes.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Logistics and Supply-Chain Foundations

Use the concept flow to revise observation, principle, method and verification.

Module 2: Warehousing, Inventory, Transportation and Distribution

Use the concept flow to revise observation, principle, method and verification.

Module 3: Planning, Sourcing, Coordination and Performance

Use the concept flow to revise observation, principle, method and verification.

Module 4: Digital, Sustainable and Resilient Supply Chains

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Map material, information and cash flows for a simple manufactured product and identify three hand-off risks.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare two warehouse/distribution alternatives using a transparent service–cost–risk–emissions table.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate a simple educational inventory or forecast example, clearly documenting assumptions and limitations.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Design a balanced supply-chain KPI dashboard for a case organisation, including data source and review frequency.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Create a reverse-logistics process map that includes quality, safety, data and sustainability considerations.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Logistics and Supply-Chain Foundations

Explain From logistics to supply-chain management and state one correct method, application or precaution.−

Logistics coordinates movement, storage and related information; supply-chain management coordinates a wider network of organisations and decisions from sourcing through operations, distribution and customer service. Performance depends on connected flows rather than one function in isolation.

Answer framework: Map a simple product journey from supplier to customer and label material, information and cash flows, hand-offs, lead times, ownership and likely service risks.

Important point: A simplified map may omit compliance, labour, safety, finance and contractual conditions. State scope and assumptions before using it for analysis.

Explain Stakeholders, objectives and trade-offs and state one correct method, application or precaution.−

Customers, suppliers, manufacturers, service providers, regulators and communities can have different priorities. Decisions balance service level, cost, speed, reliability, capacity, risk, quality and environmental/social impact.

Answer framework: Use a stakeholder-and-objective table to identify a trade-off in a case, the metric each stakeholder values and the evidence needed before deciding.

Important point: Do not optimise a local metric such as transport cost if it transfers unacceptable risk, delay, waste or service failure elsewhere in the chain.

Module 2 — Warehousing, Inventory, Transportation and Distribution

Explain Warehousing and inventory fundamentals and state one correct

method, application or precaution.−

Warehouses receive, store, protect, pick and dispatch goods while supporting consolidation, postponement and service availability. Inventory protects against uncertainty and supports flow, but it ties up capital and can create obsolescence, loss or handling risk.

Answer framework: Classify stock in a case (cycle, safety, seasonal or pipeline), identify relevant costs and data, then explain how lead time and demand variability influence the inventory policy.

Important point: EOQ, reorder-point and safety-stock examples rely on assumptions. Use actual demand, service, shelf-life, capacity and policy data before operational use.

Explain Transport and distribution-network design and state one correct method, application or precaution.−

Transport modes, consolidation, routes, distribution-centre placement and last-mile arrangements affect cost, speed, capacity, emissions and resilience. Distribution networks must align with product characteristics and customer-service needs.

Answer framework: Compare two distribution alternatives with a transparent table covering distance, mode, lead time, handling points, capacity, service level, risk, emissions and information gaps.

Important point: Routing and facility recommendations need accurate network, vehicle, labour, regulatory and safety data; educational cases are not dispatch instructions.

Module 3 — Planning, Sourcing, Coordination and Performance

Explain Forecasting and operational planning and state one correct method, application or precaution.−

Forecasts inform capacity, procurement, inventory and distribution plans but remain uncertain. Planning links demand pattern, lead time, capacity, constraints and service targets; it benefits from regular review and feedback.

Answer framework: Create a simple forecast-review worksheet: historical demand source, method assumption, forecast, error check, event adjustment, capacity implication and decision owner.

Important point: Forecasts are estimates, not facts. Avoid using one short data series or an untested model as the sole basis for commitments or emergency stock decisions.

Explain Sourcing, coordination and performance and state one correct method, application or precaution.−

Make-or-buy, supplier selection, vendor rating and coordination affect continuity, cost, quality and risk. Useful KPIs balance service, cost, quality, assets, flexibility and sustainability rather than rewarding one silo alone.

Answer framework: Propose a small KPI dashboard for a case, defining each metric, calculation boundary, source, frequency, target rationale and unintended behaviour to watch for.

Important point: Supplier scoring and contracts may involve confidential data, fairness, legal duties and strategic risk. Use authorised data and review processes.

Module 4 — Digital, Sustainable and Resilient Supply Chains

Explain Digitalisation and visibility and state one correct method, application or precaution.−

Digital tools can improve visibility, planning, traceability and coordination when data quality, interoperability, cybersecurity and user workflows are addressed. Technology does not remove the need for governance and exception management.

Answer framework: Map a data flow for an order or shipment and identify data owner, update point, decision user, quality risk, access control and recovery action for missing/incorrect data.

Important point: Operational data can be commercially sensitive and security-critical. Do not share, connect or automate systems without authorisation and appropriate controls.

Explain Sustainability, reverse logistics and resilience and state one correct method, application or precaution.−

Green logistics seeks lower environmental impact; reverse logistics manages returns, recovery, repair, reuse or disposal; resilience prepares networks to absorb and recover from disruption. These choices require a whole-life, whole-network view.

Answer framework: For a product-return case, identify return triggers, triage options, safety/quality checks, recovery route, transport/packaging needs, data capture and measures of environmental/service impact.

Important point: Never assume returned goods are safe or reusable. Regulated, hazardous, damaged or personal-data-bearing goods need authorised handling and disposition processes.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Logistics and Supply-Chain Foundations.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Warehousing, Inventory, Transportation and Distribution.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Planning, Sourcing, Coordination and Performance.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Digital, Sustainable and Resilient Supply Chains.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Logistics definition and evolution; SCM networks; product, information and financial flows; stakeholders; service, cost and value; manufacturing and service contexts..
2. Develop a complete answer or practical plan for: Storage and warehouse roles; inventory costs and policies; transportation modes and routing; distribution

networks, facility location and service design..
3. Develop a complete answer or practical plan for: Demand forecasting, aggregate/capacity planning, scheduling awareness, make-or-buy, sourcing and vendor rating, coordination, KPIs and balanced performance views..
4. Develop a complete answer or practical plan for: E-logistics, traceability and data; reverse, green and lean logistics; disruption risk, resilience, emergency response, circularity and responsible asset management..

LAST-MILE REVISION

Quick Revision

Module 1: Logistics and Supply-Chain Foundations

Logistics definition and evolution; SCM networks; product, information and financial flows; stakeholders; service, cost and value; manufacturing and service contexts.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Warehousing, Inventory, Transportation and Distribution

Storage and warehouse roles; inventory costs and policies; transportation modes and routing; distribution networks, facility location and service design.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Planning, Sourcing, Coordination and Performance

Demand forecasting, aggregate/capacity planning, scheduling awareness, make-or-buy, sourcing and vendor rating, coordination, KPIs and balanced performance views.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Digital, Sustainable and Resilient Supply Chains

E-logistics, traceability and data; reverse, green and lean logistics; disruption risk, resilience, emergency response, circularity and responsible asset management.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
From logistics to supply-chain management	Logistics coordinates movement, storage and related information; supply-chain management coordinates a wider network of organisations and decisions from sourcing through operations, distribution and customer service.
Stakeholders, objectives and trade-offs	Customers, suppliers, manufacturers, service providers, regulators and communities can have different priorities.
Warehousing and inventory fundamentals	Warehouses receive, store, protect, pick and dispatch goods while supporting consolidation, postponement and service availability.
Transport and distribution-network design	Transport modes, consolidation, routes, distribution-centre placement and last-mile arrangements affect cost, speed, capacity, emissions and resilience.
Forecasting and operational planning	Forecasts inform capacity, procurement, inventory and distribution plans but remain uncertain.
Sourcing, coordination and performance	Make-or-buy, supplier selection, vendor rating and coordination affect continuity, cost, quality and risk.
Digitalisation and visibility	Digital tools can improve visibility, planning, traceability and coordination when data quality, interoperability, cybersecurity and user workflows are addressed.
Sustainability, reverse logistics and resilience	Green logistics seeks lower environmental impact; reverse logistics manages returns, recovery, repair, reuse or disposal; resilience prepares networks to absorb and recover from disruption.

LEARNING RESOURCES

References

Recommended logistics and supply-chain references

1. Sunil Chopra, Supply Chain Management: Strategy, Planning, and Operation.
2. Donald J. Bowersox et al., Supply Chain Logistics Management.
3. Martin Christopher, Logistics & Supply Chain Management.
4. R. Panneerselvam, Operations Research.

Approved public logistics and supply-chain sources

- https://nptel.ac.in/courses/109105494
- https://nptel.ac.in/courses/110106045

These sources provide the verified study basis while official Course 6023A detail is unavailable; they do not replace contracts, transport regulations, safety requirements, customs rules or an organisation’s approved operational data.